

Thilina Hemaka Weerakkody, Ph.D.

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Research Interests

- **Human–Autonomy Teaming and Cognitive Systems Engineering:** Human-centered control, shared autonomy, and decision-support systems for robotics and safety-critical applications.
- **Advanced Control of Nonlinear and Uncertain Systems:** Robust and adaptive control (including sampled-data and $\mathcal{L}_1$ architectures), real-time optimization, and stability analysis for nonlinear robotic and actuator systems.
- **Embodied AI and Intelligent Robotics:** Algorithms for autonomous behavior, motion planning, learning-enhanced control, and embodied decision-making in robotic systems.
- **Physics-Informed and Data-Driven Modeling:** Integration of physics-based models with machine learning for prediction, control, and system identification in uncertain environments.
- **Real-Time and Embedded Robotic Systems:** ROS-based sensing and control architectures, real-time implementation, and software/hardware integration for experimental robotic platforms.

Education

2019–2024	PhD in Mechanical Engineering, University of Iowa, IA. <i>Thesis: Design and Control of Artificial Muscles for Robotic Applications</i> <i>Advisors: Dr. Caterina Lamuta and Dr. Venzio Cichella</i>
2011–2016	B.Sc. (Hons.) in Mechanical Engineering, University of Moratuwa, Sri Lanka. <i>Second Class-Upper Division</i> <i>Thesis: Development and Control of a Modular Artificial Shoulder Prosthetic Device.</i>
2012–2014	Diploma in Information Technology (HEQ), British Computer Society (BCS).

Employment History

Postdoctoral Scholar, CNSI, University of California Los Angeles, CA. **2024–Present**

Advisor: Prof. R. Michael van Dam

Research focus: High-throughput radiochemistry robotic platform development and optimization, In situ radioactivity measurements in high-throughput radiochemistry.

Research Assistant, SMMS Lab, University of Iowa, IA. **2019–2024**

Advisors: Dr. Caterina Lamuta and Dr. Venzio Cichella

Research focus: Novel mechanism design and development, Mathematical and control system modeling, Underwater, rehabilitation, and medical robotic system development, and real-time control.

Graduate Teaching Assistant, Mechanical Engineering, University of Iowa, IA. **2023–2024**

ME6130: Novel Artificial Muscles & Sensors, *Tutorial/Lab session on MATLAB/Simulink for mathematical modeling in graduate-level research. (Spring 2023-2024)*

Research Engineer, Juggernaut Life Sciences – Funded by NIH, Iowa City, IA. 2020–2023
External collaborator with PhD advisor; responsible for CAD modeling (SOLIDWORKS/Fusion 360), novel mechanism design, and mechanical & control system development for medical robot applications.

Instructor (Assistant Lecturer), Mechanical Engineering, SLIIT, Sri Lanka. 2018–2019
ME2021: Mechanics of Machines, ME2031: Engineering Drawing, ME3531: Solid Mechanics & Mechanical Design, ME3620: Control Systems, ME2541/ME3640: Mechatronics System.

Mechatronics Lab Manager, Mechanical Engineering, SLIIT, Sri Lanka. 2018–2018
Projects: Developed embedded system lab setups for undergraduate Mechanical/Mechatronics courses.

Research Assistant, Bionics Lab, Univ. of Moratuwa, Sri Lanka. 2016–2018
Research focus: Designing lower-limb prostheses and an adaptive foot device considering kinematics, dynamics, and 3D print prototyping (Funded by National Research Council grant 15-068).

Graduate Teaching Assistant, Mechanical Eng., Univ. of Moratuwa, Sri Lanka. 2016–2018
ME4700/ME4310: COMSOL multi-physics structural modeling for MEMS/NEMS, ME1802: Introduction to Manufacturing Engineering, ME2160: Automotive transmission system & brake system, ME2040: Fundamentals of Mechatronics, ME2023: Manufacturing Engineering I, ME4462: Automation Systems, TT4162: Control Systems & Applications, ME5124: Automation and Control of Manufacturing Systems (Graduate-level course).

Engineering Intern, Shin Nippon Air Technologies Co. Ltd., Sri Lanka. 2014–2015
Refrigeration & Air system designing. On-site MEP Supervision at Aitken Spence Resort Project.

Technical Skills

Programming & Software:	Python, C/C++, MATLAB, Simulink, LabVIEW, ROS1/ROS2, Git/GitHub, Linux (Ubuntu), Docker.
Embedded Systems:	Arduino, Raspberry Pi, Teensy, embedded C/C++; SPI, I2C, UART; motor drivers, custom electronics, real-time control architectures.
Control & Modeling:	Robust/adaptive control, sampled-data control, model-based control, non-linear dynamics, physics-based modeling; MATLAB Control Toolbox; NumPy, SciPy.
Machine Learning & Vision:	Python ML stack (NumPy/SciPy, scikit-learn), PyTorch (intermediate), TensorFlow (beginner), OpenCV (project-level); physics-informed modeling and learning-based control.
Simulation & Planning:	Gazebo (Classic and Ignition), RViz, MoveIt, MuJoCo, PyBullet, URDF/Xacro modeling, ROS-based visualization and motion planning.
CAD, Mechanics & FEA:	SolidWorks, Fusion 360, AutoCAD, COMSOL, ABAQUS; mechanism design, kinematics, and dynamics.
Sensing & Actuation:	Load cells, thermocouples, encoders, EMG (Delsys), optical motion capture, camera-based imaging; signal conditioning.
Fabrication & Prototyping:	3D-printing (FDM/SLA), molding, soldering, and rapid prototyping.
PCB Design:	EasyEDA, Altium (project-level).

Certifications

1. Certificate in Teaching Methodology in Higher Education conducted by Sri Lanka Institute of Information Technology (SLIIT) (August - December 2018).

Awards

2023	Editor's Choice Article in MDPI - Robotics .
2016	Research assistant fellowship, National Research Council, SL. (Grant (15-068))
2016	Dean's List–Semester 8, Undergraduate studies, University of Moratuwa, Sri Lanka.

Professional Membership

2012 – Present	Institute of Electrical and Electronic Engineers (IEEE)–92445858
2016 – Present	Associate Engineer, Institution of Engineers in Sri Lanka, (IESL)
2015 – Present	IEEE Robotics and Automation Society
2018 – Present	IEEE Control Systems Society
2019 – Present	Registered Practitioner, Engineering Council Sri Lanka, (ECSL)–206823
2024 – Present	American Society of Mechanical Engineers, (ASME)–000103812978
2024 – Present	American Institute of Aeronautics and Astronautics, (AIAA)
2024 – Present	Society of Industrial and Applied Mathematics (SIAM)

Patents

1. Muhammad Taifur Rahman, Nashwaan A. Khan, Mahmudul Alam Shakib, **Thilina H. Weerakkody**, Ibrahim Razu, and Marlan Hansen, “Semi-Automated Immunolabeling Systems and Associated Devices and Methods”, U.S. Patent Application [US 2025/0231210 A1](#), published 2025/07/17.
2. **Pending** - Automated Sample Preparation Device (ASAP Device). *This innovation was designed in collaboration with the Department of Otolaryngology at the University of Iowa to automate the sample preparation process for Immunolabeling. (**Disclosed** with the University of Iowa Research Foundation (UIRF))*

Peer-review Publications

Journals

1. **Thilina H. Weerakkody**, Elio Matteo Curcio, Giuseppe Carbone, Carmine Maletta, Emanuele Sgambitterra, and Caterina Lamuta, “Robust Control of Shape Memory Alloys for Assistive Robotics Applications”, *Journal of Shape Memory and Superelasticity* (**Accepted**).
2. Sean Maxson, **Thilina H. Weerakkody**, Maxwell Hammond, Venanzio Cichella, and Caterina Lamuta, “Design of a low-cost continuum soft robotic arm powered by Twisted and Coiled Artificial Muscles (TCAMs)”, *ASME Journal of Mechanisms and Robotics* (**Under-review**).
3. Rabiul Mamman, **Thilina H. Weerakkody**, and Caterina Lamuta, “Bioinspired Active Vortex Generators for Underwater Flow Control”, *Robotics Reports*, 3(1), 26-36, 2025.
4. Rabiul Mamman, Tatum Johnson, **Thilina H. Weerakkody**, and Caterina Lamuta, “Fouling Release Mechanism of an Octopus-Inspired Smart Skin”, *Advanced Functional Materials*, p. 2406405, 2024.
5. Sean Maxson, Parth Kotak, **Thilina H. Weerakkody**, and Caterina Lamuta, “Manufacturing and underwater cyclic behavior of different types of twisted and coiled artificial muscles (TCAMs)”, *Manufacturing Letters*, Elsevier, 2024.
6. Parth Kotak, Sean Maxson, **Thilina H. Weerakkody**, Venanzio Cichella, and Caterina Lamuta, “Octopus-Inspired Muscular Hydrostats Powered By Twisted and Coiled Artificial Muscles”, *Soft Robotics*, 2023.
7. Rabiul Mamman, Parth Kotak, **Thilina H. Weerakkody**, Tatum Johnson, Austin Krebill, James Buchholz, and Caterina Lamuta, “Deployable vortex generators for low-Reynolds-number applications powered by cephalopod-inspired artificial muscles”, *iScience*, 26(12), 2023.

8. **Thilina H. Weerakkody**, Maxwell Hammond, James H Neilan, Venanzio Cichella, and Caterina Lamuta, “Modeling and control of twisted and coiled artificial muscles for Soft Robotics”, *Meccanica*, 58(4):643–658, 2023.
9. Carlo Greco, **Thilina H. Weerakkody**, Venanzio Cichella, Leonardo Pagnotta, and Caterina Lamuta, “Lightweight Bioinspired Exoskeleton for Wrist Rehabilitation Powered by Twisted and Coiled Artificial Muscles”, *Robotics*, 12(1):27, 2023. ([Editor’s Choice Article](#))
10. Samantha Bell, Arnold Bangel, **Thilina Weerakkody**, Xuan Song, and Caterina Lamuta, “Automated manufacturing system for carbon fiber-based twisted and coiled artificial muscles (TCAMs)”, *Manufacturing Letters*, 33:19–23, 2022.
11. Maxwell Hammond, Venanzio Cichella, **Thilina H. Weerakkody**, and Caterina Lamuta, “Robust and adaptive sampled-data control of twisted and coiled artificial muscles”, *IEEE Control Systems Letters*, 6:1232–1237, 2021.
12. Parth Kotak, **Thilina H. Weerakkody**, and Caterina Lamuta, “Physics-based dynamic model for the electro-thermal actuation of bio-inspired twisted spiral artificial muscles”, *Polymer*, 222:123642, 2021.
13. **Thilina H. Weerakkody**, Thilina Dulantha Lalitharatne, and R.A.R.C. Gopura, “Adaptive foot in lower-limb prostheses”, *Journal of Robotics*, 2017.

Conference Proceedings

1. Rabiul Mamman, **Thilina H. Weerakkody**, and Caterina Lamuta, “Bioinspired Active Vortex Generators for Enhanced Underwater Flow Control,” in *ASME 2025 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*, 2025.
2. Mahmudul A. Shakib, Zhaolin Gao, **Thilina H. Weerakkody**, Aditya L. Tamilkovan, and Caterina Lamuta, “Geopolymers for Neuromorphic and Piezoelectric Applications,” in *ASME 2024 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*, 2024.
3. Elio M. Curcio, **Thilina H. Weerakkody**, Emanuele Sgambitterra, Carmine Maletta, Giuseppe Carbone, Venanzio Cichella, Caterina Lamuta, “Adaptive control strategy for SMA-based artificial muscles,” in ESOMAT (13th European Symposium on Martensitic Transformation), Lecco, Italy, 26-30 August, 2024.
4. Sean Maxson, Parth Kotak, **Thilina H. Weerakkody**, and Caterina Lamuta, “Soft Tentacles for Underwater Robotics Powered by Twisted and Coiled Artificial Muscles (TCAMs),” in *ASME SMASIS 2023 Austin, TX*, 11-13 September 2023.
5. **Thilina H. Weerakkody**, Maxwell Hammond, Venanzio Cichella, and Caterina Lamuta, “Dynamic Modelling and Robust Control for Twisted and Coiled Artificial Muscles,” in *ASME SMASIS 2022 Dearborn, MI*, 12-14 September 2022.
6. Maxwell Hammond, Venanzio Cichella, **Thilina H. Weerakkody**, and Caterina Lamuta, “Robust and Adaptive Sampled-Data Control of Twisted and Coiled Artificial Muscles,” in *2021 IEEE Conference on Decision and Control*, Dec. 2021.
7. **Thilina H. Weerakkody**, Parth Kotak, Mahmudul A. Shakib, and Caterina Lamuta, “Cephalopod-inspired muscular hydrostats from twisted and coiled artificial muscles (TCAMs),” in *ASME 2021 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*, 2021.
8. **Thilina H. Weerakkody**, Parth Kotak, and Caterina Lamuta, “Artificial papillae for self-morphing skin: A Dynamic Model,” in *Society of Engineering Science*, 2020.
9. Parth Kotak, **Thilina H. Weerakkody**, Casey Harwood, James Buchholz, and Caterina Lamuta, “Boundary Layer Transition Induced by Twisted Spiral Artificial Muscles (TSAMs),” in *ASME 2020 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*, 2020.
10. Carlo Greco, **Thilina H. Weerakkody**, Calvin Kielas-Jensen, Venanzio Cichella, Leonardo Pagnotta and Caterina Lamuta, “Lightweight and Anthropomorphic Assistive Robotics from Twisted and Coiled Artificial Muscles,” in *Society of Engineering Science (SES) Conference*, 2020.

11. Carlo Greco, **Thilina H. Weerakkody**, Calvin Kielas-Jensen, Venanzio Cichella, Leonardo Pagnotta, and Caterina Lamuta, “Rehabilitation Glove powered by Twisted and Coiled Artificial Muscles,” in *ASME SMASIS* 2020.
12. **Thilina H. Weerakkody**, Nirmala N. Liyanaarachchi, H.M. Chinthaka Herath, R.A.R.C. Gopura, and Thilina D. Lalitharatne, “Development of an Active Shoulder Prosthesis with Low-Level Control Validation,” In *Proceedings of the IASTED International Conference Modeling, Identification and Control*, volume 848, pages 194–199, 2017, ACTA Press.

Poster Presentations

1. George Elias, Braeden Harrell, **Thilina H. Weerakkody**, Kirsten M. Anderson, Jason Wilken, Caterina Lamuta, Deema Totah, “Adaptive ankle foot orthoses stiffness powered by artificial muscles”, American Society of Biomechanics, Madison, Wisconsin, Aug 5–8, 2024.
2. Sean Maxson, **Thilina H. Weerakkody**, Caterina Lamuta, “Design of a low-cost continuum soft robotic tentacle powered by Twisted and Coiled Artificial Muscles (TCAMs)”, Research Open House, CoE UIowa, April 25, 2024.
3. Rabiul Mamman, Tatum Johnson, **Thilina H. Weerakkody**, Caterina Lamuta, “Theoretical model for antifouling performance of bio-inspired smart skin”, Research Open House, CoE UIowa, April 25, 2024.
4. George Elias, Marissa McFadden, **Thilina H. Weerakkody**, Braeden Harrell, Kirsten M. Anderson, Jason Wilken, Caterina Lamuta, Deema Totah, “Ankle Foot Orthoses Powered Using Artificial Muscles”, Research Open House, CoE UIowa, April 25, 2024.
5. Sean Maxson, Parth Kotak, **Thilina H. Weerakkody**, Caterina Lamuta, “Octopus-inspired Soft tentacles”, Research Open House, CoE UIowa, April 20, 2023.
6. **Thilina H. Weerakkody**, Maxwell Hammond, Venanzio Cichella, Caterina Lamuta, “Twisted and Coiled Artificial Muscles Modeling and Control”, Research Open House, CoE UIowa, April 15, 2022.
7. **Thilina H. Weerakkody**, Carlo Greco, Venanzio Cichella, Leonardo Pagnotta, Caterina Lamuta, “Wearable rehabilitation glove powered by Twisted and Coiled Artificial Muscles (TCAMs)” in Research Open House 2020.

Talks

1. “Design and Control of Artificial Muscles for Robotic Applications”, at the Mechanical Engineering Graduate Seminar at the University of Iowa, February 29, 2024.

Review Experience

1. IEEE International Conference on Systems, Man, and Cybernetics (SMC), Oct. 2017, Banff, Canada.
2. 6th IEEE Region 10 Humanitarian Technology Conference 2018 (R10 HTC’18), Dec. 2018, Sri Lanka.
3. [ASME - The Journal of Medical Devices](#) Reviewer, 2025 – present.
4. [ASME - The Journal of Mechanisms and Robotics](#) Reviewer, 2025 – present.
5. [MERCon 2025 - Moratuwa Engineering Research Conference](#), Aug. 14-15, 2025, Sri Lanka.

News Coverage

1. Iowa researchers’ octopus-like design improves underwater vehicle maneuverability [Iowa Now](#).
2. Rehabilitation glove, The University of Iowa: [The Daily Iowan](#), [UIowa-engineering news](#).
3. Robot Battle 2018–Uwa Wellassa University, Sri Lanka: [UWU Link](#), [YouTube Link](#).
4. SAKURA Science Exchange Program participation (2018), Sri Lanka. [Daily Mirror](#), [UOM Link](#)

Competitions

- **IEEE Xtream Programming Competition** — Participated in editions 6.0 (2012), 7.0 (2013), and 8.0 (2014).

Exchange Program Representation

1. Representing Sri Lanka for Japan-Asia Youth Exchange Program in Science (SAKURA) held in January/February 2018, Miyazaki, Japan. Attended the International Conference on Artificial Life and Robotics 2018, held in Beppu, Oita, Japan, from February 1-4, 2018.

Volunteering Experience

2018	Adjudicator-Panelist, Robot Battle 2018 , Uwa Wellassa University, Sri Lanka.
2017	Exhibitor, ExMo 2017 , University of Moratuwa, Sri Lanka.
2016	Exhibitor, Techno 2016 , Organized by IESL Sri Lanka.
2015 – 2015	Assistant Treasurer, IMechE (UK) Student Chapter , University of Moratuwa, SL.
2016 – 2022	Rotaract Club of Alumni of University of Moratuwa, Sri Lanka.
	2017/18: Vice President .
	2016/17: Community Service Director.
2012 – 2015	Rotaract Club of University of Moratuwa, Sri Lanka.